has already been installed for transmission of voice and data. Optical disk recording with semi conductor, laser playback are replacing the conventional piezo electric pickups.

Various optical materials and their Application

Technology	Application
1. Optical communication	
i) Wave guides	To guide the light inside the fibre by total internal reflection (eg. optic fibre cable)
ii) Optical modulator	To modulate the intensity or phase of light by an electric field / magnetic field /ultrasonic waves
iii) Optical switches	To do fast switching
iv) Optical Source	To produce light
2. Energy convertors	To convert light energy into electrical energy (solar panel, silicon, selenium sulphate)
3 Thermal energy detectors	To detect thermal radiation
4 Display device	To display electrical signal in the visual (eg LED, LASER)
5 Optical Fiber Sensor	To measure mechanical quantities like displacement, acceleration, pressure & electrical quantities likes field strength, current

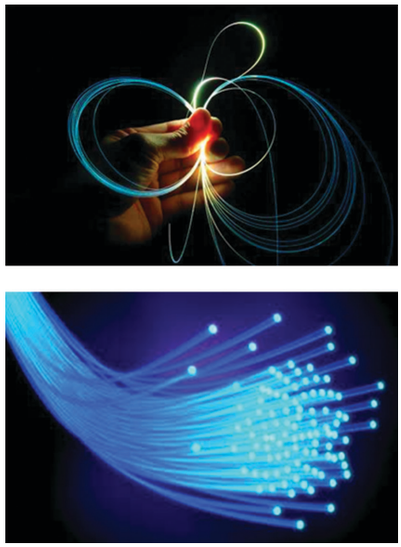


Fig 9.7 *Examples of Optical Fibres*

Activities

1. Collect 15 insulating Materials?

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